The contingent value of process information in online manufacturing time prediction*

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Abstract

Online prediction of when a job will complete is critical for shop floor planning, customer communication, and intervention. Models can rely on the aggregate state of the system (queueing models, time series) or case-level trajectories using event logs to estimate job outcomes. Prior work has overlooked how characteristics of the manufacturing process affect the incremental accuracy gain of case-based over state-based predictions. Using discrete event simulation with configurable parameters of a manufacturing process, and analyzing synthetic event data, we find this gain is smallest in processes with low path entropy and low between-path time variance. We propose a contingency view that links the value of case-level data to quantified process structure.

Keywords

online time prediction, predictive process monitoring, manufacturing, operations management

1. Introduction

Lead time prediction is a prominent problem in manufacturing operations management, as it is critical for production planning, customer communication, and process improvement. Traditionally, this is a problem approached from an apriori perspective, producing a single estimate of total cycle time or time of completion at job arrival (). The classical OR approach rely on the state of the system, meaning how congested the system is relative to its capacity, to estimate how long jobs will take. Once a prediction is made and the job is released into the shop floor, no new prediction is made. As an industry heuristic, remaining time is derived by subtracting elapsed time from the initial forecast, or by summing the average historical run times of the remaining stages.

The increasing availability of detailed event log data from manufacturing execution systems creates both the opportunity and the pressure to move beyond this static approach. This is particularly timely as manufacturers transition toward more flexible manufacturing systems (Weckenborg et al., 2024), where longer, more complex production processes with increasingly probabilistic routing make a single apriori prediction insufficient — and where the job's unfolding path becomes a critical source of predictive information. (CLAUDE's proofread need to change later)

Predictive process monitoring (PPM) is a newer approach that emerged within the field of process mining that can address this problem. Rather than relying on system-wide state, PPM conditions each prediction on the individual job's observed history — the sequence of activities and machines visited, the timestamp of each completed activity, and the elapsed time accumulated so far. The core assumption is that future outcomes will resemble those of historical jobs with similar histories. Because a new prediction can be issued after every completed activity, PPM models update dynamically as the job progresses, which is a property that static, state-based forecasts do not share.

Still, these path-aware models represent significant investment in data handling, information systems

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and modelling capabilities. This raises an important question for both OR and PPM researchers: Under what circumstances does case-path information significantly elevate predictive performance? Conversely, when do state-based approaches offer adequate accuracy?

In this study, we tackle this problem from an intuitive lens: the structural characteristics of the manufacturing process itself. Specifically, we study how two measurements of a process' structure influence the informational value of path knowledge (knowing all the possible paths and being able to infer the remaining route of the job), and therefore, determine the conditions under which case-path prediction has the biggest advantage over the traditional state-based approach.

The first measurement is between-path variance, representing how much of total cycle time variance is explained by the choice of path, and the second measurement is entropy, representing the frequency distribution of paths across all observed jobs. Together, these two dimensions define a contingency framework that links the structural properties of a process to the predictive value of case-path information.

Our study establishes and provides first evidence for this contingency view. Using discrete event simulation with configurable process parameters, we generate synthetic event logs spanning a wide range of structural configurations and compare the predictive accuracy of a process-aware Flow Analysis model against two state-based baselines — Simple Exponential Smoothing and Little's Law formula. We find that while the case-path model outperforms on all datasets, the magnitude of this advantage varies substantially and predictably with process structure. Specifically, we find that both between-path variance and path entropy have a positive and statistically significant relationship with the relative accuracy gain of the case-path approach, and that there is an interaction effect between the two dimensions: entropy matters more when variance is high, and vice versa. Together, the two dimensions explain 68% of the variation in relative accuracy across 84 synthetic logs. (CLAUDE)

Our paper's value lies in addressing the two tensions. The first is theoretical: As PPM studies increasingly incorporate queueing-related features into ML predictors, and as OM researchers gain access to richer event-level process data, understanding the conditions under which each type of information matters most becomes essential for designing efficient and effective prediction systems. Our study provides first evidence that process structure is a meaningful lens to look at this question and our proposed measurements, between-path variance and path entropy, are useful indicators of the contingent value of the event-level process data. The second tension is practical: practitioners investing in event log infrastructure and PPM capabilities have no way of knowing in advance whether the accuracy gain will justify the cost. Our findings suggest that this question has a structural answer, one that managers can assess by measuring observable characteristics of their own process before committing to the investment.

The remainder of this paper is organized as follows. Section 2 reviews the literature on state-based and case-path approaches to lead time prediction and motivates the contingency view. Section 3 introduces the theoretical framework and defines the two process characteristics and the relative accuracy metric. Section 4 describes the simulation methodology and experimental design. Section 5 presents the results. Section 6 discusses the findings and their implications, and Section 7 concludes.

2. Literature Review

2.1. The State-based approach to Lead Time Prediction problem in Manufacturing OR

Queueing Theory is the study of queues...Especially important in manufacturing where the majority of cycle time comes from waiting in queues rather than actual processing.

Time Series Analysis is a versatile method used in a wide variety of predictive problems, including cycle times () and wait times () in manufacturing systems. Its core focus is on temporal patterns and trends.

2.2. The case-path alternative

Process mining is a discipline developed at the intersection of business process management and data science.

It analyzses event logs - digital records produced by information systems that capture the execution of business processes (van der Aalst, 2016)

Table 1
A minimal event log example

Case ID	Activity	Timestamp	Resource
Job 01	Moulding - end	2024-01-10 08:03	Machine A
Job 01	Assembly - end	2024-01-10 11:47	Machine C
Job 01	Packaging - end	2024-01-10 15:22	Machine E
Job 02	Moulding - end	2024-01-10 08:51	Machine A
Job 02	Assembly - end	2024-01-10 13:10	Machine D
Job 02	Packaging - end	2024-01-10 17:45	Machine F

From event logs, process mining methods automatically generate models that accurately represent how business processes are actually executed, as opposed to how they are expected to be executed (process discovery). These models are then used to check whether execution conforms to the standard model (conformance checking), enhance process performance, and predict the future or outcome of an going process. It is this last branch -specifically the subfield of predictive process monitoring, that is relevant for our case-path prediction approach.

Predictive Process Monitoring aims at predicting the future of an ongoing (uncompleted) process execution (job). Typical examples of predictions include categorical outcome, the sequence of the next activities, or completion time of the job (Francescomarino and Ghidini, 2022). In this study, we focus on remaining time prediction. The high-level implementation workflow is similar to that of a general ML problem, as it consists of 2 phases: a training or learning phase, where a predictive model is learned from logs of historical (complete) jobs. and a runtime/prediction phase, where the model is queried for predicting the future of an ongoing case (Francescomarino and Ghidini, 2022) using its prefix - the partial trace of acitivities, timestamps, and contextual attributes recorded for the case up to the prediction point. The core (simplified) assumption is that future completion/remaining time will resemble those of historical jobs whose prefixes were similar: jobs that visited the same sequence of stations, consumed similar time at each stage, or uoperated under similar conditions are expected to complete in similar ways. PPM apporaches can be divided into generative and discriminative models (). Generative models explicitly reconstruct the process structure, estimating path probabilities and each possible path's completion times, and use the observed prefix to infer the most likely route. Discriminative models treat prediction as regressional ML task, mapping prefix features directly to an outcome without modelling the process explicitly; some prominent models include (). Additional features, including system-wide and local state features can be leveraged. However, the case-level prefix is prerequisite to make any prediction at all. As such, PPM algorithms are able to produce a new of the remaining time or completion time of an ongoing job after completion of each stage, incorporating its most recent progresses. The ability to dynamically reflect on the job's own routing history and update its prediction means the case-path approach supposedly provides better accuracy than the more static, state-based predictors.

2.3. The tension and motivation for our study

Both the state-base OM and the case-path based PPM models can leverage different computation tools and can be enriched with a wide range of domain-specific covariates, like job and shop characteristics, and doing so generally improves their accuracy. However, the primary mechanisms are fundamentally different: the former relies on the state, or the momentum of the system; while the latter conditions on the incomplete history of the target case itself. Due to this structural difference, state-based and case-path based methods have been studied separately (). To provide a middle ground, we study the potential of PPM's case-path view to improve the dynamic time predictions beyond the state-based view. But more than that, we want to find the boundaries of the value-add of this case-path information. Most time prediction studies perform mass evaluation to find the most consistently outperformning model(s), without asking "under what circumstances", or how these model's relative accuracy changes across different kinds of dataset or process. We see this as a missed opportunity to deepen our understanding of the technologies that exist, and use that understanding to fuel both effective application as well as development of new technologies that address the caveats of the existing ones.

It was natural to suspect that predictive accuracy depends not only on model sophistication or data richness, but also on the characteristics of the underlying system. Yet in the context of multi-stage manufacturing processes, we are not aware of any study systematically addressing the relationship between the characteristics of the process itself and the effectiveness of a time prediction model. Some studies have explored the boundaries of existing predictors for single-stage service systems. Bassamboo and Ibrahim (2020) seek to move beyond mass evaluation and develop a contingency view to explain the relative perfromance of different prediction methods. Specifically, they compare the accuracy of static (average waiting times over a long horizon) against the LES snapshot predictor- which reports the delay of the last customer to have entered service at the moment a new customer arrives and is never updated thereafter. They show that the LES predictor outperforms the static announcement if and only if the correlation between LES predictions and actual waiting times exceeds 1/2. Chocron et al. 2022 explored the conditions under which traditional Queueing Theory-based predictors are preferred over machine learning algorithms by varying by testing both approaches across systems that increasingly violate the clean assumptions queueing theory relies on — such as exponential service times, stationary arrival rates, and a first-come-first-served discipline. They find that ML algorithms outperform when FCFS priority discipline is violated.

Our study shares the same motivation but differs from these studies in two ways. The first difference is that, while both Bassamboo and Ibrahim (2020) and Chocron (2022) study single stage service queues and assume a single prediction is made at the moment of arrival, we investigate multi-stage manufacturing processes. In this setting, aggregate state-based approaches behave much like the predictors in those earlier studies: a prediction is made when a job arrives and is never genuinely revised — as the job progresses, elapsed time is simply subtracted from the initial forecast, with no new information about the job itself incorporated. Process-aware approaches work differently. After every completed activity, they revise their prediction based on the specific route the job has taken so far, using that accumulated case history to narrow down what is likely to happen next. The two approaches therefore draw on fundamentally different sources of information: aggregate approaches ask what the system looks like right now, while process-aware approaches ask what has happened to this particular job so far. (CLADUE TEXT need to change).

This first difference motivates the second, which concerns our contingency criterion. Rather than properties of a single queue, such as statisical correlation between the prediction and the observation, arrival rates and service times distribution, or priority rules violation, as in Bassamboo and Ibrahim (2020) and Chocron et al. (2022), we focus on the structure of the process: how many unique paths jobs can take, how much faster or slower some paths are compared to others, and how jobs are distributed across these paths. Although we utilize statiscal methods to quantify these characteristics using historical data, the metrics computed essentially represents the physical configuration of the process rather than the temporal fluctuations of the queue. In the methodological framework that follows, we introduce their definitions and explain how we examine their relationship with the relative accuracy of the two approaches, state-based and case-path based.

3. Theoretical framework

We try to address the research question by first asking: *What structural characteristics of a process make path knowledge beneficial?* Naturally, there are two main characteristics.

3.1. Explained variance by path choice

First: *How much of cycle time variance can be explained by different choices of path?* This characteristic can be measured as the ratio of between-path variance to total variance:

$$D_{\text{path}} = \frac{\text{Var}_{\text{between}}}{\text{Var}_{\text{total}}} \tag{1}$$

Where:

$$\text{Var}_{\text{total}} = \frac{1}{N} \sum_{i=1}^{N} (CT_i - \overline{CT})^2 \tag{2}$$

$$\text{Var}_{\text{between}} = \sum_{p=1}^{K} \pi_p (\overline{CT}_p - \overline{CT})^2 \tag{3}$$

$$\text{Var}_p = \frac{1}{n_p} \sum_{j=1}^{n_p} (CT_j - \overline{CT}_p)^2 \tag{4}$$

3.2. Distribution of paths

Second: *Does the process have few dominant paths, or evenly distributed paths?* To measure this characteristic, we use Shannon entropy, a concept from Information Theory (Shannon,1948):

$$H = -\sum_{p=1}^{K} \pi_p \log \pi_p \tag{5}$$

We normalize entropy to make it comparable across different numbers of paths:

$$H_{\text{norm}} = \begin{cases} 0 & \text{if } K = 1 \\ \frac{H}{\log K} & \text{if } K > 1 \end{cases} \tag{6}$$

Interpretation:

- $H_{\text{norm}} = 0$: all cases follow a single path
- $H_{\text{norm}} = 1$: all paths are equally likely

We use the natural logarithm in computing entropy. Since H_{norm} is normalized by $\log K$, the choice of logarithmic base does not affect the results.

3.3. Notation

Intuitively, the performance gap between knowing and not knowing the path is widest in processes with high D_{path} and low H_{norm}.

3.4. Relative Accuracy Gain (RAG)

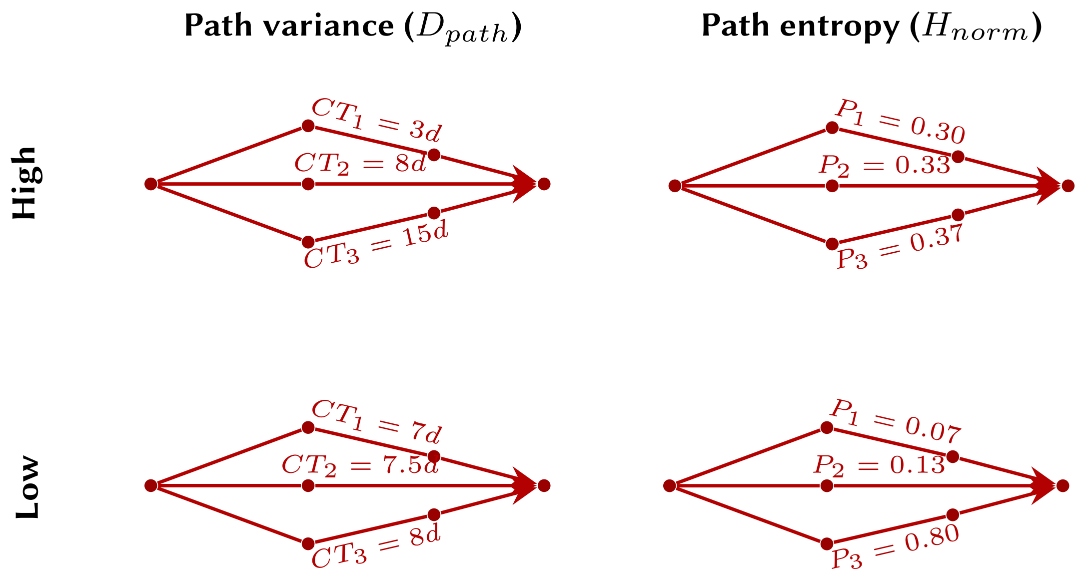
The next problem is defining and measuring this performance gap. MAE (Mean Absolute Error) is a common evaluation metric used for benchmarking in PPM studies. The limitation of MAE is that it is scale-depedent, meaning it is not reliable for comparison across different datasets. We therefore use MAE to create a scale-independent metric, which we call the Relative Accuracy Gain (RAG):

$$\text{RAG} = \frac{MAE_{\text{FA}} - \min(MAE_{\text{SES}}, MAE_{\text{QT}})}{MAE_{\text{FA}}} \tag{7}$$

Symbol	Description
N	Total number of cases
K	Number of distinct machine-level paths
n_p	Number of cases using path p
$\pi_p = \frac{n_p}{N}$	Probability (share) of path p
p	Path index, $1 \leq p \leq K$
CT_i	Cycle time of case i
$\overline{CT}$	Mean cycle time of all cases
$\overline{CT}_p$	Mean cycle time of path p
Var_p	Variance of cycle time within path p
H	Shannon entropy

Table 2
Notation used in the theoretical framework

Figure 1: Illustrative examples of D_{path} and H_{norm} behavior



Where the Mean Absolute Error (MAE) for each model is defined as:

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} |R_i - \widehat{R}_i| \tag{8}$$

Notation:

N : Total number of events evaluated in the test set

i : Event index

R_i : Actual remaining time

$\widehat{R}_i$: Predicted remaining time

The RAG compares the performance of the process-aware model (FA) against the best-performing system-state baseline (SES or QT). The interpretation of the metric is as follows:

- **Positive RAG**: Indicates that the FA (Case view) model is more accurate.
- **Negative RAG**: Indicates that the SES or QT (System-state view) models are more accurate.

4. Methodology

Let us recall that the main focus of the study is to investigate the relationship between process characteristics and the predictive performance. This is often hard to do with empirical data, because different real-life event logs come from vastly different settings and industries, with varying sample sizes, available attributes, quantities, etc., not just process complexity, making it difficult to isolate the impact of structural process characteristics on the final performance. We overcome this limitation by using controlled simulation to compare different event logs with varying levels of process structural complexity while still keeping the general setup.

We assume a manufacturing setup where the machines are equipped with readers that produce only completion timestamps for each activity. If a dataset has both start and completion timestamps for each event, one can also estimate the queue lengths and wait time at each machine, however this is not a realistic setup for most organizations, and is also not necessary to establish standard prefix data.

For both SES and Little's Law, we implement the predictors with a top down approach, meaning we predict the total cycle time at the beginning of the job, using timestamps of only the first and the last actvities, lock this prediction in, only deriving remaining time from subtracting elapsed time. This reflects the classical apriori OM approach and implementation is simple.

For Flow Analysis, we leverage also the data from the intermediary stations. However, since only completion timestamps are available, the predictor only knows system-wide WIP, not machine-level congestion, and no other job or shop chateristics are used as input features. The only information advantage FA has over the state-based baselines is the case-path data.

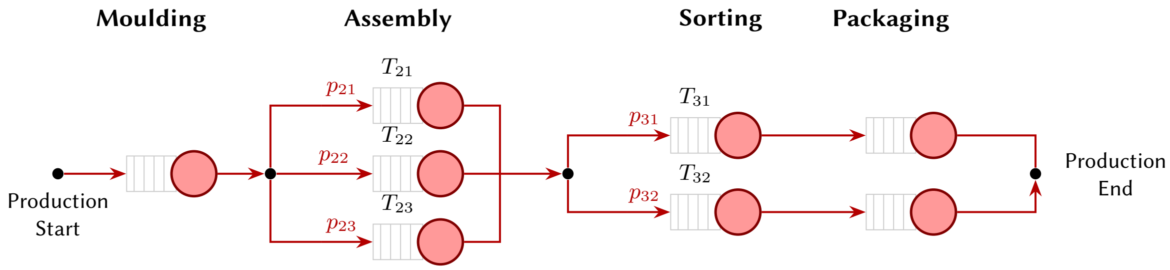
4.1. Discrete Event Simulation

The purpose of the DES is to generate synthetic event logs datasets used for implementation of the predictors. We developed a simulation framework that builds upon the source code from Busch & Leopold (2026). Specifically, the general object model for processes, tasks and resources, as well as basic event-scheduling loop was reused. The process structures, task sets, and routing logic were reconfigured to represent a multi-station manufacturing line with quality control and rework, inspired by Lorenz et al. (2021). The fixed assumptions include:

- An open, single-class Jackson network
- Poisson arrivals with average rate $\lambda = 0.28$ per simulation time unit. This means that in different simulation runs of length 1000, there might be approximately 270 or 290 jobs but on average across many runs, the average will be around 280 jobs.
- exponential and independent service times across all stations
- No machine breakdown
- Machines are assigned through a First Come First Served (FCFS) Policy
- Unlimited queue length
- No buffer inventory
- Transport time between stations on the conveyor system is not logged, assuming immediate transfer, meaning all time is spent either waiting in queues or being processed on machines.
- Run time is 30000*

At the activity lelvel, the process follows a fixed sequential production line layout, where every job visits every stage in the same order (Papadopoulos and Heavey;198X, 1996) and follows the same path. In our study, however, we define paths at the machine level: two jobs completing the same stage on

Figure 2: Illustrative manufacturing scenario: stages and parameters.



different machines are considered to be on distinct paths. This machine-level definition creates Dpath and Hnorm variation in a fixed sequence production line, while reflecting the granularity often seen in equipped manufacturing systems. A second source of path variation is probabislitic rework: jobs failting quality control re-enter that stage's queue, extending their machine-level trace as well as cycle time.

Table 3
Experimental factors, their levels, and theoretical justification

Factor	Levels	Why
Queue style	• Pooled: machines share one queue • Dedicated: each machine has its own queue • Hybrid	Dedicated routing creates more distinct path combinations, leading to higher H_{norm}.
QC Pass Rate	1.0, 0.99, 0.97, 0.95, 0.90	Lower pass rates introduce rework loops, increasing H_{norm} and creating cycle time variance between rework vs. non-rework cases.
# of activities (stations)	5, 8, 12	Additional stations create more branching points (higher H_{norm}) and accumulate more path-based cycle time variances (higher D_{path}).
Machine Heterogeneity	• Identical: Equal service rates • Strong: Service speeds differ greatly • Mild: Slight speed differences	Differences in server speed create fixed cycle time differences between specific paths, resulting in higher D_{path}.

For each job, the simulator records the completion times for each activity, the used resources and a timestamp.

The generated logs are then preprocessed into prefix data for making predictions using the following procedure:

To avoid initialization bias, the first 17% of cases were trimmed off, along with any incomplete case by the end of runtime. To generate a wide range of {D_path} and {H_norm} values across the datasets, the pipeline allows for experimenting with the following factors.

4.2. Title variants

5. Results

We assume a manufacturing setup where the machines are equipped with readers that produce only completion timestamps for each activity. If a dataset has both start and completion timestamps for each event, one can also estimate the queue lengths and wait time at each machine, however this is not a realistic setup for most organizations, and is also not necessary to establish standard prefix data. From these timestamps, three information sources can be derived: 1) Historical cycle times of all past cases 2) System-wide WIP - by counting open jobs (started but not completed) 3) Case-specific prefix - including the sequence of stations and resources visited and elapsed time for each job. For both SES and Little's Law, we implement the predictors with a top down approach, meaning we predict the total cycle time at the beginning of the job, using timestamps of only the first and the last actvities, lock this prediction in, only deriving remaining time from subtracting elapsed time. This reflects the classical apriori OM approach and implementation is simple.

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6. Discussion

We assume a manufacturing setup where the machines are equipped with readers that produce only completion timestamps for each activity. If a dataset has both start and completion timestamps for each event, one can also estimate the queue lengths and wait time at each machine, however this is not a realistic setup for most organizations, and is also not necessary to establish standard prefix data.

For both SES and Little's Law, we implement the predictors with a top down approach, meaning we predict the total cycle time at the beginning of the job, using timestamps of only the first and the last actvities, lock this prediction in, only deriving remaining time from subtracting elapsed time. This reflects the classical apriori OM approach as well as the information capabilities of most manufacturers, and the implementation is simple.

For Flow Analysis, we leverage also the data from the intermediary stations. However, since only completion timestamps are available, the predictor only knows system-wide WIP, not machine-level congestion, and no other job or shop chateristics are used as input features. The only information advantage FA has over the state-based baselines is the case-path data.

7. Conclusion

We thereby introduce a contingency view that links process characteristics to predictive value of two different information types/approaches. This study directly addresses lead time prediction problems in multi-stage processes with probabilistic routing where all jobs are visible, which is an unique characteristic of manufacturing job shops, online retail fulfillment ETA. However, the relevance extends further. Consider a service system where the prediction target is the wait time for a single queue (e.g., patients waiting to start treatment). This appears to be a one-stage problem. But the service itself (the treatment) may consist of multiple internal stages whose durations vary systematically with patient characteristics and path. By improving the prediction of this internal multi-stage service duration, our method indirectly improves the accuracy of the external queue wait time prediction. The total time from arrival to departure is the sum of queue wait and service time; better prediction of the latter benefits the former. Thus, even problems that present as single-stage can benefit from this research whenever the underlying service has decomposable structure.

Another limitation of this study is that we stop the analysis at predictive accuracy without translating it to actual economic costs of either approaches. This is an area where we believe the practitioner-the owner of the process, can bridge the gap, as they understand what their process looks like, what the purpose of the prediction is, and also the specific tangible costs associated with accuracy loss and IT capital investment. As such, our study serves as the initial evidence for the presence as well as the contingent magnitude of the simplicity-accuracy tradeoff they are making.

Table 4
Frequency of Special Characters

Non-English or Math	Frequency	Comments
Ø	1 in 1,000	For Swedish names
π	1 in 5	Common in math
$	4 in 5	Used in business
Ψ^2_1	1 in 40,000	Unexplained usage

Table 5
Some Typical Commands

Command	A Number	Comments
`\author`	100	Author
`\table`	300	For tables
`\table*`	400	For wider tables

7.1. Authors and Affiliations

8. Sectioning Commands

Your work should use standard LaTeX sectioning commands: `\section`, `\subsection`, `\subsubsection`, and `\paragraph`. They should be numbered; do not remove the numbering from the commands.

Simulating a sectioning command by setting the first word or words of a paragraph in boldface or italicized text is not allowed.

9. Tables

The "`ceurart`" document class includes the "`booktabs`" package — https://ctan.org/pkg/booktabs — for preparing high-quality tables.

Table captions are placed *above* the table.

Because tables cannot be split across pages, the best placement for them is typically the top of the page nearest their initial cite. To ensure this proper "floating" placement of tables, use the environment `table` to enclose the table's contents and the table caption. The contents of the table itself must go in the `tabular` environment, to be aligned properly in rows and columns, with the desired horizontal and vertical rules.

Immediately following this sentence is the point at which Table 4 is included in the input file; compare the placement of the table here with the table in the printed output of this document.

To set a wider table, which takes up the whole width of the page's live area, use the environment `table*` to enclose the table's contents and the table caption. As with a single-column table, this wide table will "float" to a location deemed more desirable. Immediately following this sentence is the point at which Table 5 is included in the input file; again, it is instructive to compare the placement of the table here with the table in the printed output of this document.

10. Math Equations

You may want to display math equations in three distinct styles: inline, numbered or non-numbered display. Each of the three are discussed in the next sections.

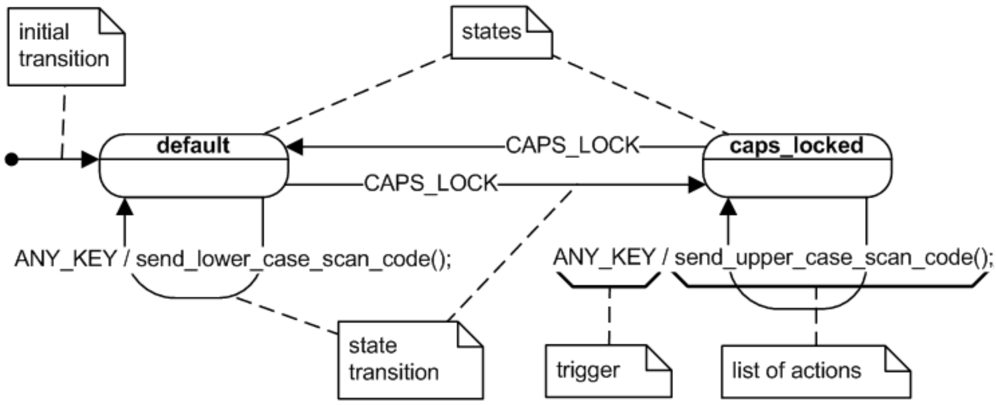

Figure 3: Sample UML state machine (source: https://en.wikipedia.org/wiki/UML_state_machine, license CC BY-SA 3.0).

10.1. Inline (In-text) Equations

A formula that appears in the running text is called an inline or in-text formula. It is produced by the `math` environment, which can be invoked with the usual `\begin` ... `\end` construction or with the short form `$` ... `$`. You can use any of the symbols and structures, from α to ω, available in LaTeX [1]; this section will simply show a few examples of in-text equations in context. Notice how this equation: $\lim_{n\rightarrow \infty} \frac{1}{n} = 0$, set here in in-line math style, looks slightly different when set in display style. (See next section).

10.2. Display Equations

A numbered display equation—one set off by vertical space from the text and centered horizontally—is produced by the `equation` environment. An unnumbered display equation is produced by the `displaymath` environment.

Again, in either environment, you can use any of the symbols and structures available in LaTeX; this section will just give a couple of examples of display equations in context. First, consider the equation, shown as an inline equation above:

$$\lim_{n\rightarrow \infty} \frac{1}{n} = 0. \tag{9}$$

Notice how it is formatted somewhat differently in the `displaymath` environment. Now, we'll enter an unnumbered equation:

$$S_n = \sum_{i=1}^{n} x_i,$$

and follow it with another numbered equation:

$$\lim_{x\rightarrow 0} (1+x)^{1/x} = e \tag{10}$$

just to demonstrate LaTeX's able handling of numbering.

11. Figures

The "`figure`" environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.

Your figures should contain a caption which describes the figure to the reader. Figure captions go below the figure. Your figures should also include a description suitable for screen readers, to assist the visually-challenged to better understand your work.

Figure captions are placed below the figure.

12. Citations and Bibliographies

The use of BibTeX for the preparation and formatting of one's references is strongly recommended. Authors' names should be complete — use full first names ("Donald E. Knuth") not initials ("D. E. Knuth") — and the salient identifying features of a reference should be included: title, year, volume, number, pages, article DOI, etc.

The bibliography is included in your source document with these two commands, placed just before the `\end{document}` command:

```
\bibliography{bibfile}
```

where "`bibfile`" is the name, without the "`.bib`" suffix, of the BibTeX file.

12.1. Some examples

A paginated journal article [2], an enumerated journal article [3], a reference to an entire issue [4], a monograph (whole book) [5], a monograph/whole book in a series (see 2a in spec. document) [6], a divisible-book such as an anthology or compilation [7] followed by the same example, however we only output the series if the volume number is given [8] (so series should not be present since it has no vol. no.), a chapter in a divisible book [9], a chapter in a divisible book in a series [10], a multi-volume work as book [11], an article in a proceedings (of a conference, symposium, workshop for example) (paginated proceedings article) [12], a proceedings article with all possible elements [13], an example of an enumerated proceedings article [14], an informally published work [15], a doctoral dissertation [16], a master's thesis: [17], an online document / world wide web resource [18, 19, 20], a video game (Case 1) [21] and (Case 2) [22] and [23] and (Case 3) a patent [24], work accepted for publication [25], prolific author [26] and [27]. Other cites might contain 'duplicate' DOI and URLs (some SIAM articles) [28]. Multi-volume works as books [29] and [30]. A couple of citations with DOIs: [31, 28]. Online citations: [32, 18, 33, 34].

13. Acknowledgments

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document.

This section has a special environment:

```
\begin{acknowledgments}
  These are different acknowledgments.
\end{acknowledgments}
```

so that the information contained therein can be more easily collected during the article metadata extraction phase, and to ensure consistency in the spelling of the section heading.

Authors should not prepare this section as a numbered or unnumbered `\section`; please use the "`acknowledgments`" environment.

14. Appendices

If your work needs an appendix, add it before the "`\end{document}`" command at the conclusion of your source document.

Start the appendix with the "`\appendix`" command:

```
\appendix
```

and note that in the appendix, sections are lettered, not numbered.

Acknowledgments

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Declaration on Generative AI

Either:
The author(s) have not employed any Generative AI tools.

Or (by using the activity taxonomy in ceur-ws.org/genai-tax.html):
During the preparation of this work, the author(s) used X-GPT-4 and Gramby in order to: Grammar and spelling check. Further, the author(s) used X-AI-IMG for figures 3 and 4 in order to: Generate images. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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A. Online Resources

The sources for the ceur-art style are available via

- GitHub,
- Overleaf template.